

Поиск файлов и фильтрация данных

Презентация по лабораторной работе №11

true

today

Докладчик

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Цель работы

Познакомиться с текстовым редактором Emacs. Получить практические навыки работы с Emacs в операционной системе Linux, включая создание, редактирование, сохранение файлов, а также управление буферами и окнами.

Задание

- Запустить редактор Emacs
- Создать файл lab07.sh
- Ввести заданный текст
- Сохранить файл
- Выполнить операции редактирования (вырезание, копирование, вставка, отмена)
- Освоить команды перемещения курсора
- Изучить работу с буферами
- Освоить управление окнами
- Выполнить поиск и замену текста

Emacs — это мощный многофункциональный текстовый редактор, используемый в системах Linux и Unix. Он отличается высокой расширяемостью и использованием комбинаций клавиш для выполнения команд.

Основные возможности Emacs: - редактирование текстовых файлов

- работа с несколькими буферами
- разделение окна на несколько областей
- выполнение поиска и замены текста
- поддержка различных режимов

В отличие от редактора vi, Emacs не использует режимы работы, а все команды выполняются через сочетания клавиш (Ctrl, Alt).

Выполнение лабораторной работы

Задание 1. Работа в редакторе Emacs

0.установка Emacs

```
libotf x86_64 VBoxClient: the VirtualBox kernel
libtree-sitter x86_64 service is not running. Exiting.
3
3
становка слабых зависимостей:
info x86_64 7.2-7.fc43 updates 353.9 KiB
libtree-sitter-bash x86_64 0.25.1-1.fc43 updates 1.3 MiB
libtree-sitter-c x86_64 0.24.1-3.fc43 fedora 614.9 KiB
libtree-sitter-c-sharp x86_64 0.23.1-2.fc43 fedora 5.7 MiB
libtree-sitter-cmake x86_64 0.7.1-2.fc43 fedora 99.7 KiB
libtree-sitter-cpp x86_64 0.23.4-5.fc43 fedora 3.3 MiB
libtree-sitter-css x86_64 0.23.2-2.fc43 fedora 109.6 KiB
libtree-sitter-elixir x86_64 0.3.4-4.fc43 fedora 1.4 MiB
libtree-sitter-go x86_64 0.25.0-1.fc43 fedora 227.6 KiB
libtree-sitter-heap x86_64 0.8.0-2.fc43 fedora 33.8 KiB
libtree-sitter-html x86_64 0.23.2-2.fc43 fedora 33.4 KiB
libtree-sitter-java x86_64 0.23.5-5.fc43 fedora 416.6 KiB
libtree-sitter-javascript x86_64 0.23.1-2.fc43 fedora 362.0 KiB
libtree-sitter-jsdoc x86_64 0.23.2-2.fc43 fedora 29.1 KiB
libtree-sitter-json x86_64 0.24.8-5.fc43 fedora 17.1 KiB
libtree-sitter-lua x86_64 0.4.0-3.fc43 fedora 65.9 KiB
libtree-sitter-php x86_64 0.24.2-2.fc43 fedora 2.0 MiB
libtree-sitter-phpdoc x86_64 0.1.8-2.fc43 updates 208.8 KiB
libtree-sitter-python x86_64 0.23.6-2.fc43 fedora 468.9 KiB
libtree-sitter-ruby x86_64 0.23.1-3.fc43 fedora 2.0 MiB
libtree-sitter-rust x86_64 0.24.0-3.fc43 fedora 1.1 MiB
libtree-sitter-toml x86_64 0.7.0-3.fc43 fedora 37.2 KiB
libtree-sitter-typescript x86_64 0.23.2-4.fc43 fedora 2.8 MiB
libtree-sitter-yaml x86_64 0.7.0-3.fc43 fedora 198.6 KiB

подка транзакции:
становка: 32 пакетов

кий размер входящих пакетов составляет 90 MiB. Необходимо загрузить 90 MiB.
```

2. Создание файла

2.1 Создать файл `lab07.sh` с помощью комбинации `Ctrl + x`, `Ctrl + f`

Ввести имя файла: `lab07.sh`

```
touch lab11.sh - GNU Emacs at mka1jkhadrimokh1
```

File Edit Options Buffers Tools Sh-Script Help



```
U:--- touch lab11.sh All L1 (Shell-script[sh])
```

3. Ввод текста

3.1 Набрать следующий текст:

```
#!/bin/bash HELLO=Hello function hello { LOCAL  
HELLO=World echo $HELLO }
```


4. Сохранение файла

4.1 Сохранить файл

Ctrl + x, Ctrl + s

touch lab11.sh - GNU Emacs at mkaljkhadrimokh1

File Edit Options Buffers Tools Sh-Script Help



```
#!/bin/bash
HELL=Hello
function hello {
  LOCAL HELLO=World
  echo $HELLO
}
```

U:--- touch lab11.sh All L6 (Shell-script[sh])
(No changes need to be saved)

Задание 2. Редактирование текста

5. Основные операции

5.1 Вырезать строку (C-k)

touch lab11.sh - GNU Emacs at mka1jkhadr1mokh1

File Edit Options Buffers Tools Sh-Script Help



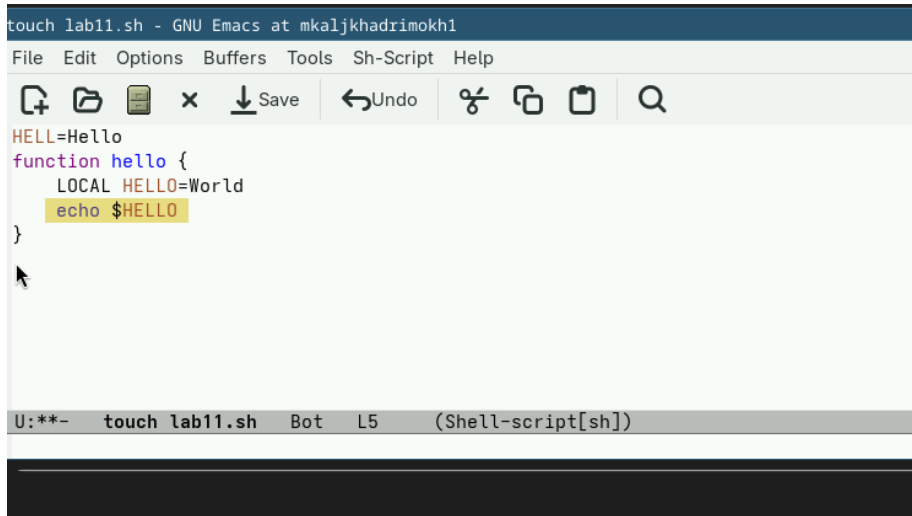
```
#!/bin/bash
HELL=Hello
function hello {
    LOCAL HELLO=World
    echo $HELLO
}
```

hello

U:**- touch lab11.sh All L8 (Shell-script[sh])

Mark set

5.3 Выделить область (C-space)



touch lab11.sh - GNU Emacs at mkaljkhadrimokh1

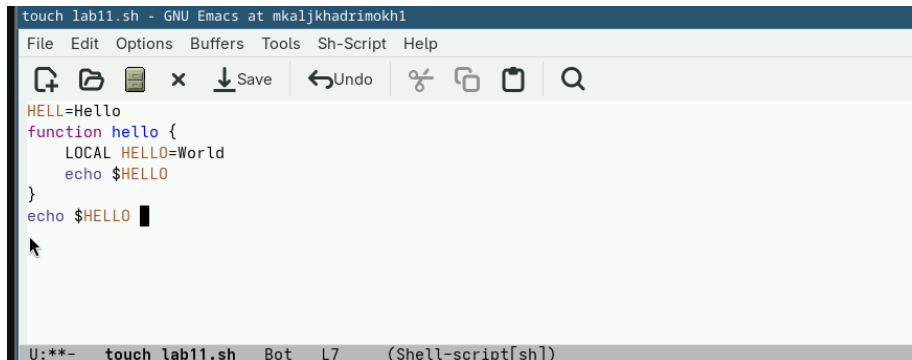
File Edit Options Buffers Tools Sh-Script Help

Icons: Copy, Paste, Save, Undo, Cut, Copy, Find

```
HELL=Hello
function hello {
    LOCAL HELLO=World
    echo $HELLO
}
```

U:*** touch lab11.sh Bot L5 (Shell-script[sh])

5.4 Скопировать область (M-w)

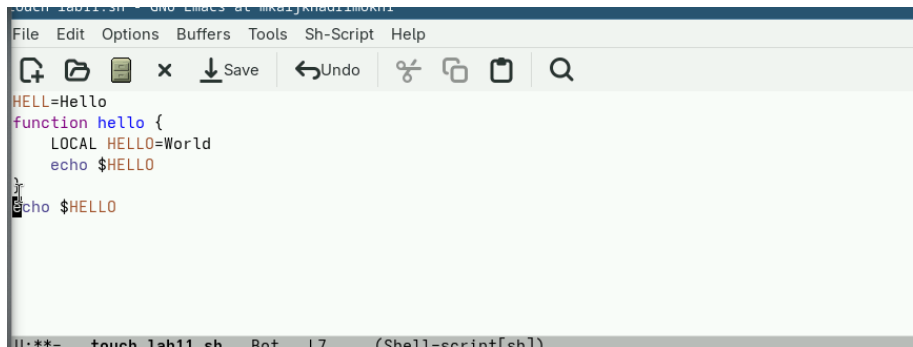


The screenshot shows the GNU Emacs editor interface. The title bar reads "touch lab11.sh - GNU Emacs at mkaljkhadrimokh1". The menu bar includes "File", "Edit", "Options", "Buffers", "Tools", "Sh-Script", and "Help". The toolbar contains icons for opening, saving, undo, redo, and search, along with text labels "Save", "Undo", and a magnifying glass. The main text area contains a shell script:

```
HELL=Hello
function hello {
    LOCAL HELLO=World
    echo $HELLO
}
echo $HELLO
```

The cursor is positioned at the end of the last line. The status bar at the bottom shows "U:***- touch lab11.sh Bot L7 (Shell-script[sh])".

5.5 Вставить область

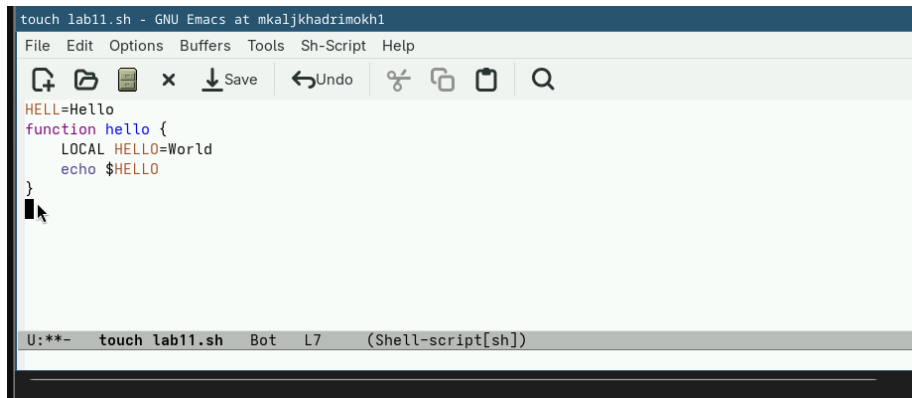


The screenshot shows a window titled "touch lab11.sh - GNU Emacs at mka1jknad1imokni". The menu bar includes "File", "Edit", "Options", "Buffers", "Tools", "Sh-Script", and "Help". The toolbar contains icons for file operations (new, open, save, close), editing (undo, redo, cut, copy, paste), and search. The script content is as follows:

```
HELL=Hello
function hello {
  LOCAL HELLO=World
  echo $HELLO
}
echo $HELLO
```

The status bar at the bottom displays "ll:**~ touch lab11.sh Bot 17 (Shell-script[sh])".

5.6 Вырезать область (C-w)

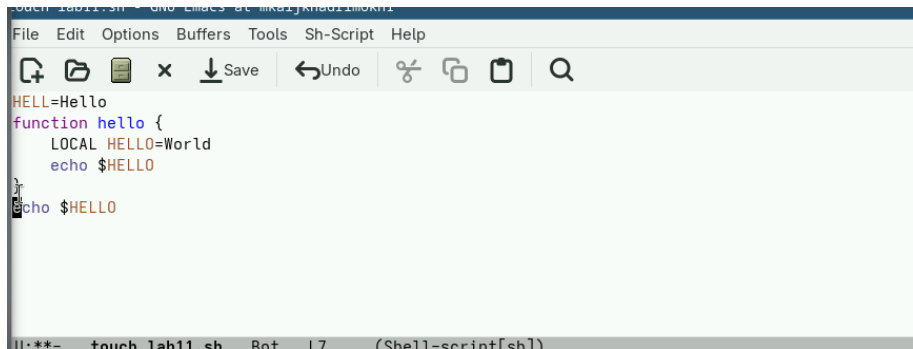


The screenshot shows the GNU Emacs editor interface. The title bar reads "touch lab11.sh - GNU Emacs at mkaljkhadrimokh1". The menu bar includes "File", "Edit", "Options", "Buffers", "Tools", "Sh-Script", and "Help". The toolbar contains icons for opening, saving, undo, redo, and search, along with labels "Save", "Undo", and a magnifying glass. The main text area contains a shell script:

```
HELL=Hello
function hello {
  LOCAL HELLO=World
  echo $HELLO
}
```

The cursor is at the end of the script. At the bottom, a terminal window shows the prompt "U:**-" followed by the command "touch lab11.sh", the output "Bot L7", and the shell name "(Shell-script[sh])".

5.7 Отмена действия (C-/)



The screenshot shows a window titled "Open lab11.sh - GNU Emacs at mka1jknad1imokni". The menu bar includes "File", "Edit", "Options", "Buffers", "Tools", "Sh-Script", and "Help". The toolbar contains icons for "Save", "Undo", "Cut", "Copy", "Paste", and "Find". The script content is as follows:

```
HELL=Hello
function hello {
  LOCAL HELLO=World
  echo $HELLO
}
echo $HELLO
```

The status bar at the bottom displays "ll:*~ - touch lab11.sh Buf 17 (Shell-script[sh])".

Задание 3. Перемещение курсора

6.1 В начало строки (C-a)

touch lab11.sh - GNU Emacs at mkaljkhadrimokh1

File Edit Options Buffers Tools Sh-Script Help



```
HELL=Hello
function hello {
  LOCAL HELLO=World
  echo $HELLO
}
echo $HELLO
```



U:***- touch lab11.sh Bot L7 (Shell-script[sh])



```
HELL=Hello
function hello {
```


6.3 В начало буфера (M-<)

touch lab11.sh - GNU Emacs at mkaljkhadrimokh1

File Edit Options Buffers Tools Sh-Script Help



Save



Undo



```
#!/bin/bash
```

```
HELL=Hello
```

```
function hello {
```

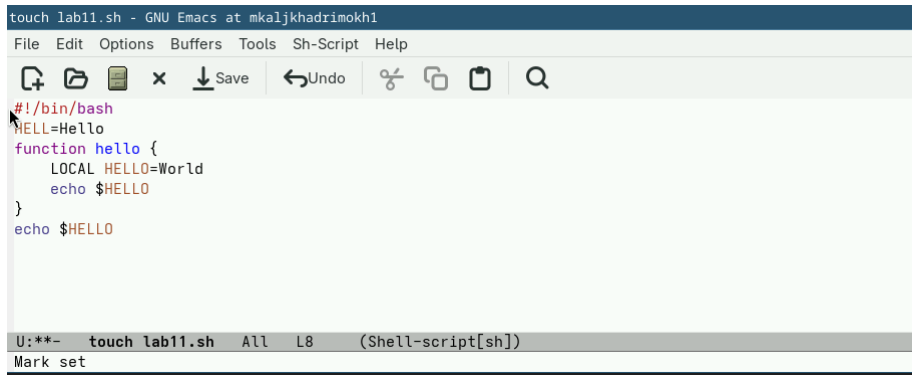
```
    LOCAL HELLO=World
```

```
    echo $HELLO
```

```
}
```

```
echo $HELLO
```

6.4 В конец буфера (M->)



```
touch lab11.sh - GNU Emacs at mkaljkhadrimokh1
File Edit Options Buffers Tools Sh-Script Help
[Icons: New, Open, Save, Close, Undo, Redo, Copy, Paste, Find]
#!/bin/bash
HELL=Hello
function hello {
    LOCAL HELLO=World
    echo $HELLO
}
echo $HELLO

U:**- touch lab11.sh All L8 (Shell-script[sh])
Mark set
```

Задание 4. Управление буферами

7.1 Список буферов (C-x C-b)

touch lab11.sh - GNU Emacs at mkaljkhadrimokh1

File Edit Options Buffers Tools Sh-Script Help



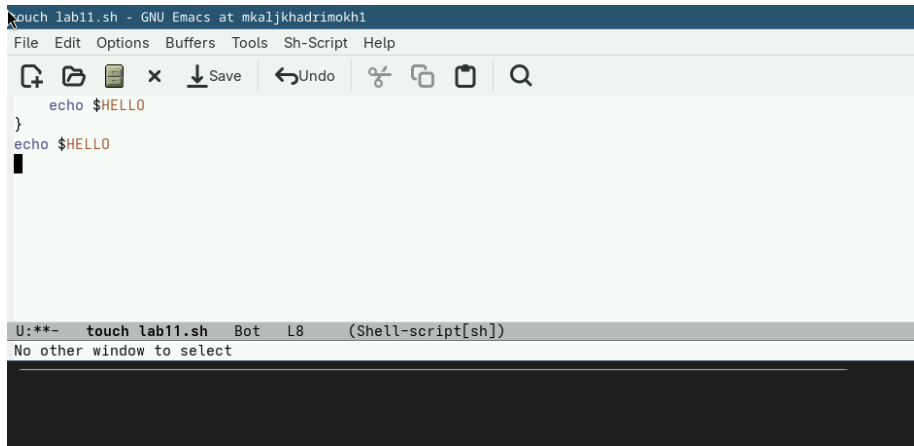
```
echo $HELLO
}
echo $HELLO
```

U:**~ touch lab11.sh Bot L8 (Shell-script[sh])

CRM Buffer	Size	Mode	File
* touch lab11.sh	94	Shell-script[sh]	~/study_2025-2026_os-intro/labs/lab11/touch lab11.sh
GNU Emacs	904	Fundamental	
scratch	147	Lisp Interaction	
Messages	1980	Messages	

U:%~ *Buffer List* All L1 (Buffer Menu)

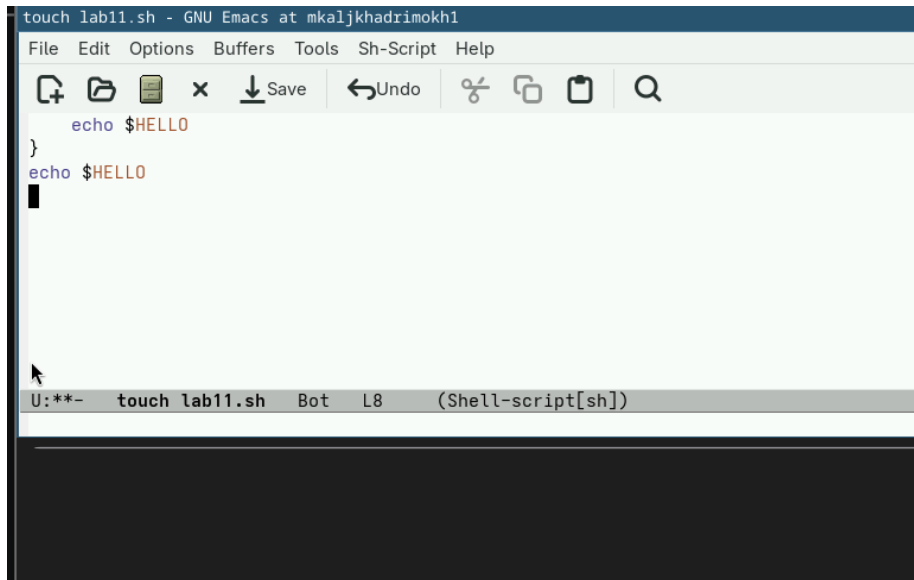
7.2 Переключение окна (C-x o)



```
touch lab11.sh - GNU Emacs at mkaljkhadrimokh1
File Edit Options Buffers Tools Sh-Script Help
[Icons: Open, Save, Undo, Copy, Paste, Find]
echo $HELLO
}
echo $HELLO
█

U:**~ touch lab11.sh Bot L8 (Shell-script[sh])
No other window to select
```

7.3 Закрытие окна (C-x 0)



```
touch lab11.sh - GNU Emacs at mkaljkhadrimokh1
```

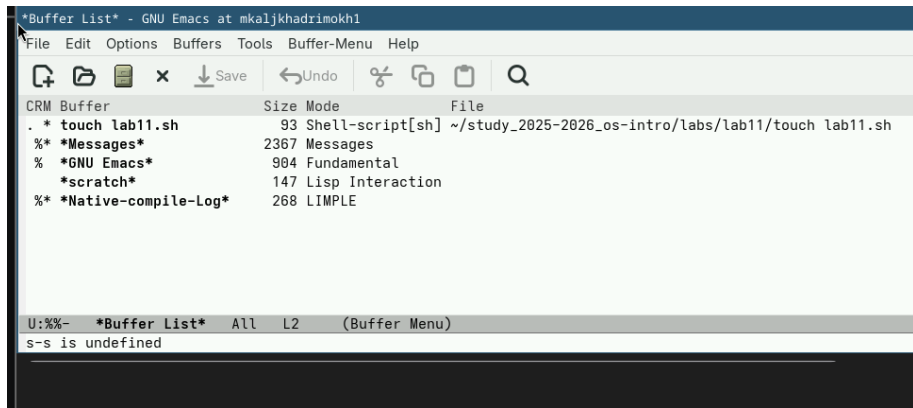
File Edit Options Buffers Tools Sh-Script Help

Save Undo

```
    echo $HELLO
}
echo $HELLO
█
```

U:**- touch lab11.sh Bot L8 (Shell-script[sh])

7.4 Переключение буферов (C-x b)



Задание 5. Управление окнами

8.1 Разделение окна

C-x 3 (вертикально)

```
touch lab11.sh - GNU Emacs at mkaljkhadrimokh1
File Edit Options Buffers Tools Sh-Script Help
[Icons: Open, Save, Undo, Cut, Copy, Paste, Find]
}
echo $HELLO
}
echo $HELLO
[Cursor]

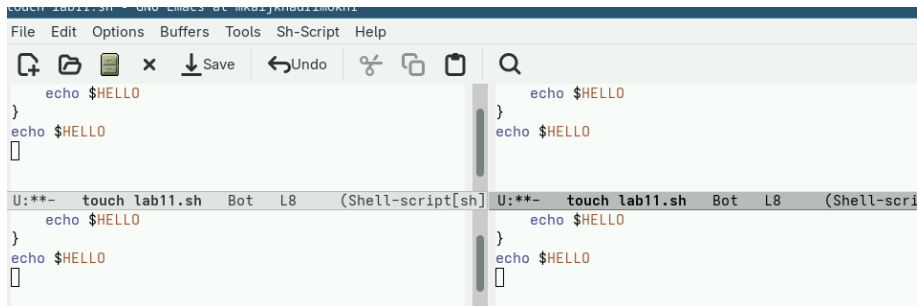
U:**- touch lab11.sh Bot L8 (Shell-script[sh]) U:**- touch lab11.sh Bot L8 (Shell-script[sh])
C-x 3
```

C-x 2 (горизонтально)

```
touch lab11.sh - GNU Emacs at mkaljkhadrimokh1
File Edit Options Buffers Tools Sh-Script Help
[Icons: Open, Save, Undo, Cut, Copy, Paste, Find]
}
echo $HELLO
}
echo $HELLO
[Cursor]

U:**- touch lab11.sh Bot L8 (Shell-script[sh]) U:**- touch lab11.sh Bot L8 (Shell-script[sh])
```

8.2 Работа в 4 окнах



Задание 6. Поиск и замена

9.1 Поиск (C-s)

The screenshot shows the GNU Emacs editor interface with a multi-buffer window. The title bar reads '*Minibuf-1* - GNU Emacs at mkaljkhadrimokh1'. The menu bar includes File, Edit, Options, Buffers, Tools, Minibuf, and Help. The toolbar contains icons for file operations (new, open, save, delete), editing (undo, redo, cut, copy, paste), and search (magnifying glass). The main editing area is split into two panes. The left pane shows a shell script snippet: `echo $HELLO`, `}`, and `echo $HELLO` with a cursor at the end of the second line. The right pane shows a similar snippet: `echo $HELLO`, `}`, and `echo $HELLO` with a cursor at the end of the second line. Below the editing area, a status bar shows the current buffer and line: `U:**-- touch lab11.sh Bot L7 (Shell-script[sh]` on the left and `U:**-- touch lab11.sh Bot L9 (Shell-scri` on the right. At the bottom, a search prompt `I-search: hell` is visible, indicating an incremental search is in progress.

9.2 Переход по результатам

The screenshot shows the GNU Emacs editor interface. The title bar reads "touch lab11.sh - GNU Emacs at mkaljkhadrimokh1". The menu bar includes "File", "Edit", "Options", "Buffers", "Tools", "Sh-Script", "Isearch", and "Help". The toolbar contains icons for navigation and editing: back, forward, repeat forward, abort, finish, undo, replace, show hits, and help. The main editing area is split into two panes. The left pane shows a shell script snippet:

```
echo $HELLO  
}  
echo $HELLO
```

 with the cursor at the end of the second line. The right pane shows the same snippet with the cursor at the end of the first line. Below the editor panes, the status bar displays search results for the term "hello". It shows four matches: "U:**- touch lab11.sh Bot L7 (Shell-script[sh]", "U:**- touch lab11.sh Bot L10 (Shell-scri", "U:**- touch lab11.sh Bot L9 (Shell-script[sh]", and "U:**- touch lab11.sh Bot L10 (Shell-scri". The first result is highlighted, and the text "Failing I-search: hello" is visible at the bottom left of the status bar.

```
touch lab11.sh - GNU Emacs at mkaljkhadrimokh1  
File Edit Options Buffers Tools Sh-Script Isearch Help  
< > Repeat Forward X Abort G Finish ↶ Undo ↷ Replace 📄 Show Hits 🛠 Help  
echo $HELLO  
}  
echo $HELLO  
U:**- touch lab11.sh Bot L7 (Shell-script[sh] U:**- touch lab11.sh Bot L10 (Shell-scri  
echo $HELLO  
}  
echo $HELLO  
U:**- touch lab11.sh Bot L9 (Shell-script[sh] U:**- touch lab11.sh Bot L10 (Shell-scri  
Failing I-search: hello
```

9.3 Выход (C-g)



A screenshot of a terminal window. The top bar shows the command 'touch lab11.sh' with a progress indicator at 65% and the label 'L8'. The terminal text shows a prompt 'U:**-' followed by the command '(Shell-script[sh] U:%*- *Message' and the word 'quit' on the next line. The cursor is positioned at the end of the 'quit' command.

```
U:**- touch lab11.sh 65% L8 (Shell-script[sh] U:%*- *Message  
quit
```

9.4 Поиск и замена (M-%)

The screenshot shows the GNU Emacs editor interface. The title bar reads '*Minibuf-1* - GNU Emacs at mkaljkhadrimokh1'. The menu bar includes File, Edit, Options, Buffers, Tools, Minibuf, and Help. The toolbar contains icons for file operations and editing. The main window displays a shell script with the following content:

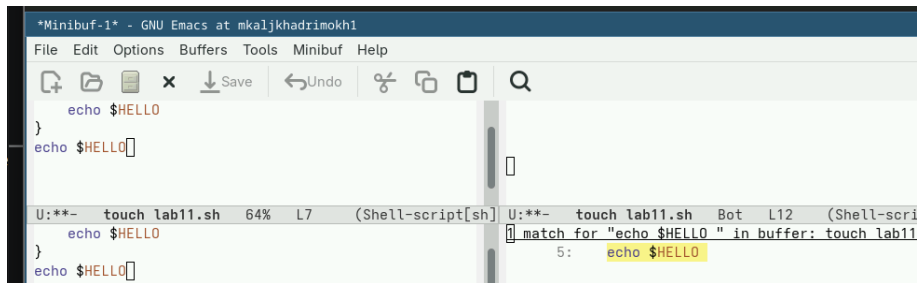
```
echo $HELLO
}
echo $HELLO
```

The cursor is at the end of the second 'echo \$HELLO' line. A status bar at the bottom shows 'U:**- touch lab11.sh 64% L7 (Shell-script[sh])'. A minibuffer window is open, displaying the search and replace operation:

```
Query replace word echo $He with: echo $HE
```

The minibuffer also shows a list of buffers on the right, including 'U:**- touch lab11.sh', 'Quit [3 times]', 'read-extended-command-1: C', 'ibuffer while in minibuffe', 'funcall-interactively: Tex', and '*Messages*'.

9.5 Альтернативный поиск (M-s o)



The screenshot shows the GNU Emacs editor interface. The title bar reads '*Minibuf-1* - GNU Emacs at mkaljkhadrimokh1'. The menu bar includes File, Edit, Options, Buffers, Tools, Minibuf, and Help. The toolbar contains icons for file operations and editing. The main text area shows a shell script snippet:

```
}  
echo $HELLO  
}  
echo $HELLO
```

The status bar at the bottom is split into two panes. The left pane shows 'U:**- touch lab11.sh 64% L7 (Shell-script[sh]'. The right pane shows 'U:**- touch lab11.sh Bot L12 (Shell-scri'. A search window is overlaid on the right pane, displaying the results of a search for 'echo \$HELLO':

```
1 match for "echo $HELLO " in buffer: touch lab11  
5: echo $HELLO
```

Выводы

В ходе выполнения лабораторной работы были получены практические навыки работы с редактором Emacs. Освоены основные команды редактирования текста, перемещения курсора, работы с буферами и окнами, а также выполнения поиска и замены.

Список литературы

- 1 Документация GNU Emacs
- 2 Учебные материалы по курсу “Архитектура компьютера и операционные системы”

